

Middle Convolution of ℓ -adic Perverse Sheaves

Submitted for the Ph.D. Candidacy Examination at Western University

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October 6, 2026

Contents

1	Introduction	2
1.1	Notations	2
1.2	Motivations	3
1.3	Acknowledgements	4
2	Local Systems and Lisse Sheaves	4
2.1	Local Systems in Topology	4
2.2	Local Systems in ℓ -adic Arithmetic Geometry	5
3	Perverse Sheaves and Middle Convolution	5
3.1	Six-Functor Formalism	5
4	Mod-ℓ Monodromy of Hyperelliptic Curves	5
5	Formalization in Lean	5

1 Introduction

1.1 Notations

1.1.1. Motivated by the growing interest in the formalization of mathematics within the mathematical community, we adopt a style of notation inspired by type theory throughout this write-up, without committing to type-theoretic foundations. The symbols below denote types and/or categories, depending on your philosophical inclination. The notation $X : T$ means that X is a term of type T or an object of the category T .

Set	Sets
Set^{fin}	Finite sets
Top	Topological spaces
$\text{Open}(T)$	Open subsets of $T : \text{Top}$
Grp	Groups
Grp^{top}	Topological groups
Grp^{pro}	Profinite groups
Grp^{ab}	Abelian groups
Ring	Rings (associative, unital, not necessarily commutative)
Ring^{top}	Topological rings
Field	Fields
$\text{Field}^{\text{top}}$	Topological fields
Mod_R	Left modules over $R : \text{Ring}$
Vect_k	Vector spaces over $k : \text{Field}$
$\text{Vect}_k^{\text{fin}}$	Finite-dimensional vector spaces over $k : \text{Field}$
$\text{Rep}_\Lambda(G)$	Continuous Λ -linear representations of $G : \text{Grp}^{\text{top}}$, where $\Lambda : \text{Ring}^{\text{top}}$
$\text{Rep}_\Lambda^{\text{fin}}(G)$	Continuous Λ -linear representations of $G : \text{Grp}^{\text{top}}$, where $\Lambda : \text{Ring}^{\text{top}}$, whose underlying topological Λ -modules are finitely generated
Sch	Schemes
Sch_S	Schemes over a base scheme $S : \text{Sch}$
Var_k	Varieties over $k : \text{Field}$, i.e. integral separated schemes of finite type over k
Curv_k	Curves over $k : \text{Field}$, i.e. one-dimensional varieties over k
$\text{Curv}_k^{\text{sm}}$	Smooth curves over $k : \text{Field}$
Riem	Riemann surfaces, i.e. connected one-dimensional complex manifolds
Site	Sites, i.e. categories equipped with Grothendieck topologies
$\text{Sh}(X, A)$	Sheaves on $X : \text{Site}$ with values in a category A admitting small limits
$\text{Mod}(X, \mathcal{R})$	Sheaves of left $\mathcal{R}$ -modules on $X : \text{Site}$, where $\mathcal{R} : \text{Sh}(X, \text{Ring})$
$D^b(A)$	Bounded derived objects of an abelian category A
$D^b(X, k)$	Bounded derived objects of $\text{Sh}(X, \text{Vect}_k)$, where $X : \text{Site}$ and $k : \text{Field}$
$X \rightarrow Y$	Morphisms from $X : T$ to $Y : T$

1.2 Motivations

1.2.1. *Monodromy groups* are interesting mathematical objects that arise naturally in both geometric and arithmetic contexts. Roughly speaking, they describe how locally defined data change when transported along a loop. Controlling and understanding monodromy groups has been a central theme in both of these worlds, with significant applications. We will discuss both the geometric and arithmetic sides, but this write-up will primarily focus on the arithmetic side, with the geometric side providing motivation and intuition.

1.2.2. Let $w : \mathbb{C}^\times$, and consider the equation

$$z^2 = w$$

Fix a base point $w_0 : \mathbb{C}^\times$ and choose $z_0 : \mathbb{C}^\times$ with $z_0^2 = w_0$. Near w_0 , the equation has two holomorphic solution branches $z_+ = \sqrt{w}$ and $z_- = -\sqrt{w}$, uniquely specified by $z_+(w_0) = z_0$ and $z_-(w_0) = -z_0$. Suppose we move w once around 0 along the loop based at w_0 given by

$$w(t) = w_0 e^{2\pi i t} \quad t : [0, 1]$$

Then we have $z_+(t) = z_0 e^{\pi i t}$ and $z_-(t) = -z_0 e^{\pi i t}$. As t goes from 0 to 1, the two solution branches swap rather than return to themselves. In other words, there is a permutation representation of the fundamental group¹ $\pi_1^{\text{top}}(\mathbb{C}^\times, w_0) \cong \mathbb{Z} : \text{Grp}^{\text{top}}$

$$\rho : \pi_1^{\text{top}}(\mathbb{C}^\times, w_0) \rightarrow \text{Aut}_{\text{Set}}(\{z_+, z_-\}) \quad n \mapsto (z_+ z_-)^n$$

We call this representation the *permutation monodromy representation* and its image, S_2 , the *permutation monodromy group* of the equation.

1.2.3. Let us reformulate the above in terms of sheaves. Consider the covering

$$f : \mathbb{C}^\times \rightarrow \mathbb{C}^\times \quad z \mapsto z^2$$

Let $\mathcal{F} : \text{Sh}(\mathbb{C}^\times, \text{Set})$ be the sheaf of sections of this cover. Thus, for $U : \text{Open}(\mathbb{C}^\times)$,

$$\begin{aligned} \mathcal{F}(U) &= \{z : U \rightarrow \mathbb{C}^\times : (f \circ z)|^U = \text{id}_U\} \\ &= \{z : U \rightarrow \mathbb{C}^\times : z(w)^2 = w, \text{ for } w : U\} \end{aligned}$$

This is a locally constant sheaf: for each $w_0 : \mathbb{C}^\times$, one can find an open neighborhood U of w_0 on which the sections are precisely the two square-root branches

$$\begin{aligned} z_+ : U &\rightarrow \mathbb{C}^\times & w &\mapsto \sqrt{w} \\ z_- : U &\rightarrow \mathbb{C}^\times & w &\mapsto -\sqrt{w} \end{aligned}$$

Hence the stalk $\mathcal{F}_{w_0} = \{z_+, z_-\} \cong f^{-1}(w_0)$ by evaluation, and therefore the aforementioned permutation representation can be rewritten as

$$\rho : \pi_1^{\text{top}}(\mathbb{C}^\times, w_0) \rightarrow \text{Aut}_{\text{Set}}(\mathcal{F}_{w_0}) \quad \gamma \mapsto [z \mapsto \tilde{\gamma}_z(1)]$$

¹with the discrete topology

where $\tilde{\gamma}_z : [0, 1] \rightarrow \mathbb{C}^\times$ is the unique lift of $\gamma : \pi_1^{\text{top}}(\mathbb{C}^\times, w_0)$ with $\tilde{\gamma}_z(0) = z$. More generally, let $\Pi_1(X)$ denote the fundamental groupoid of a topological space $X : \text{Top}$, that is, the category whose objects are points of X and whose morphisms are homotopy classes of paths, with concatenation as composition. We have a functor

$$T : \Pi_1(\mathbb{C}^\times) \rightarrow \text{Set} \quad w \mapsto \mathcal{F}_w \quad \gamma \mapsto T_\gamma$$

whose mapping on morphisms is given by the *transport map*

$$T_\gamma : \mathcal{F}_{\gamma(0)} \rightarrow \mathcal{F}_{\gamma(1)} \quad z \mapsto \tilde{\gamma}_z(1)$$

where $\tilde{\gamma}_z : [0, 1] \rightarrow \mathbb{C}^\times$ is the unique lift of $\gamma : [0, 1] \rightarrow \mathbb{C}^\times$ with $\tilde{\gamma}_z(0) = z$.

1.2.4. The sheaf $\mathcal{F}$ is locally constant but not constant: a global section would determine a germ at w_0 fixed by all monodromy, whereas transport once around 0 exchanges the two square-root germs, as demonstrated in 1.2.2. Thus $\mathcal{F}(\mathbb{C}^\times) = \emptyset$. Locally constant (but not necessarily constant) sheaves of sets, such as this one, are known as *local systems (of sets)*.

1.2.5. We can upgrade permutation monodromy representations into linear representations.

1.2.6 A Question in Arithmetic Geometry. Here is a

1.3 Acknowledgements

1.3.1. The author thanks his advisor, Prof. Chris Hall, for his guidance and advice.

1.3.2 Typesetting. The typesetting style of this write-up is inspired by [Kat88].

1.3.3 Declaration of AI Use. The use of generative AI in the preparation of this write-up was limited to the following purposes: polishing wording, phrasing, spelling, grammar, and formatting; offering writing advice; answering mathematical questions from the author to help him learn the subject; searching for and citing appropriate references; translating reading materials in French into English; retypesetting old references with archaic typesetting for ease of reading; providing emotional support for the author; providing fitness advice to help the author stay healthy and energetic while working on the write-up.

2 Local Systems and Lisse Sheaves

2.1 Local Systems in Topology

2.1.1 Notation. We use the following notation:

$X : \text{Curv}_{\mathbb{C}}^{\text{sm}}$	A smooth projective connected curve over $\mathbb{C}$
$S : \text{Set}^{\text{fin}}$	A finite nonempty subset of closed points of X
$U : \text{Curv}_{\mathbb{C}}^{\text{sm}}$	The open subvariety $X \setminus S$

$j : U \rightarrow X$	The inclusion map
$m : \mathbb{N}$	The cardinality of S
$(-)^{\text{an}} : \text{Curv}_{\mathbb{C}}^{\text{sm}} \rightarrow \text{Riem}$	The complex analytification functor

2.2 Local Systems in ℓ -adic Arithmetic Geometry

3 Perverse Sheaves and Middle Convolution

3.1 Six-Functor Formalism

4 Mod- ℓ Monodromy of Hyperelliptic Curves

5 Formalization in Lean

References

- [Kat88] Nicholas M. Katz. *Gauss Sums, Kloosterman Sums, and Monodromy Groups*, volume 116 of *Annals of Mathematics Studies*. Princeton University Press, 1988.